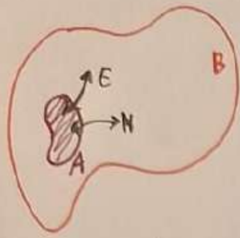
...s with angular momenta $\frac{3}{2}$ and 1, respectively, is in the state ... of $\hat{J}_i^2$ and $J_{1,2}$ $i=1,2$. Determine the values of the ...

MACROCANONICAL ENSEMBLE



$N = N_A + N_B = \text{const} \Rightarrow dN_A + dN_B = 0$

$\mu_A = \mu_B$

$dS = \frac{\partial S}{\partial N_A}\Big|_{N_B} dN_A + \frac{\partial S}{\partial N_B}\Big|_{N_A} dN_B = dN_A\left(\frac{\partial S}{\partial N_A}\Big|_{N_B} - \frac{\partial S}{\partial N_B}\Big|_{N_A}\right)$

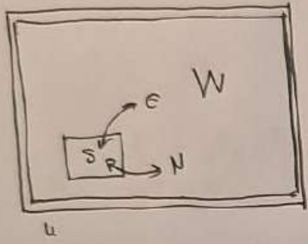
From the 1st principle of thermodynamics $dU = dQ - dW + \mu dN = TdS - pdN + \mu dN = 0$

$TdS = -\mu dN \Rightarrow \mu = -T\frac{\partial S}{\partial N}\Big|_{U,V}$

$\Rightarrow dS = dN_A \cdot \frac{1}{T}\cdot(-\mu_A + \mu_B)$

$dN_A < 0 \mapsto dS > 0$

If we have mass transfer from $A \to B$ $\mu_A > \mu_B$



$N_W >> N_S >> 1 \qquad E_W >> E_S$; $u \to NVE$; $W \to NVE$

$E_u = E_W + E_S + E_{ms} \simeq E_W \; E_S$

$$P_{m,n} = \frac{\Omega_W(E_u - E_{mn}, N_u - N; \Delta E)}{\Omega_u(E_u, N_u; \Delta E)}$$

$$\ln\Omega_W(E_W, N_W) \simeq \ln\Omega_W(E_u, N_u) + \frac{\partial \ln\Omega_W}{\partial E_W}\Big|_{\substack{E_W = E_0\\ N_W}} \overbrace{(E_W - E_u)}^{-E_{mn}} + \frac{\partial \ln\Omega_W}{\partial N_W}\Big|_{\substack{N_W = N_u\\ E_W}} \overbrace{(N_W - N_u)}^{-N}$$

$\hookrightarrow S = k\ln\Omega$

$\frac{\partial S/k}{\partial N} = \frac{1}{k}\left(-\frac{\mu}{T}\right)$

$= \ln\Omega_W(E_u, N_u) - \beta E_{mn} + \beta\mu N$

$$\Rightarrow P_{mN} = \frac{\Omega_W(E_u, N_u)\, e^{-\beta(E_{mN} - \mu N)}}{\Omega_u(E_u, N_u)} \Rightarrow 1 = \sum_{mN} P_{mN} = \frac{1}{\Xi}\sum_{mN} e^{-\beta(E_{mN} - \mu N)}$$

$\frac{1}{\Xi} \to$ Grand cannonical partition fuction

$$\Rightarrow \Xi = \sum_{m,N} e^{-\beta(E_{mN} - \mu N)}$$